

CO2-AT1-Assignment1

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QUESTIONS

Q1

Smart Document Editor using String, StringBuilder and StringBuffer

10 marks

**Q2**

Hospital Token Management System using List Interface

10 marks

**Q3**

Airport Boarding Queue Management using Queue Interface

10 marks

**Q5**

Smart Inventory Management using Map Interface

10 marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise



SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING



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5/5 answered

```
2 import java.util.Map;  
3  
4 public class Product {  
5     String name;  
6     String category;
```

```
7     double price;
8     int quantity;
9
10    public Product(String name, String category) {
11        this.name = name;
12        this.category = category;
13        this.price = price;
14        this.quantity = quantity;
15    }
16    @Override
17    public String toString() {
18        return name + " | Category: " + category;
19    }
20    public static void main(String[] args) {
21        Map<String, Product> inventory = new HashMap<>();
22
23        inventory.put("P101", new Product("Laptop", 1200, 5));
24        inventory.put("P102", new Product("Smartphone", 800, 10));
25        inventory.put("P103", new Product("Desktop", 1500, 3));
26
27        System.out.println("--- Initial Inventory ---");
28        for (Map.Entry<String, Product> entry : inventory.entrySet()) {
29            System.out.println("ID: " + entry.getKey() + " | " + entry.getValue().toString());
30        }
31
32        String updateId = "P101";
33        if (inventory.containsKey(updateId)) {
34            inventory.get(updateId).quantity++;
35            System.out.println("\nUpdated quantity for " + updateId);
36        }
37
38        String searchId = "P102";
39        System.out.println("\nSearching for " + searchId);
40        Product product = inventory.getOrDefault(searchId, new Product(searchId, 0, 0));
41        System.out.println("Found: " + product.toString());
42
43        double totalValue = 0;
44        for (Product p : inventory.values()) {
45            totalValue += p.price * p.quantity;
46        }
47        System.out.println("\nTotal Inventory Value: " + totalValue);
48
49        System.out.println("\n--- Category-wise Summary ---");
50        for (Product p : inventory.values()) {
51            if (p.category.equalsIgnoreCase("Laptop")) {
52                System.out.println("Laptop: " + p.toString());
53            } else if (p.category.equalsIgnoreCase("Smartphone")) {
54                System.out.println("Smartphone: " + p.toString());
55            } else if (p.category.equalsIgnoreCase("Desktop")) {
56                System.out.println("Desktop: " + p.toString());
57            }
58        }
59    }
60 }
```

```
52     String removeId = "P103";
53     inventory.remove(removeId);
54     System.out.println("\nRemoved discon
55
56     System.out.println("\n--- Final Avail
57     for (Map.Entry<String, Product> entr
58         System.out.println("ID: " + entr
59     }
60 }
61 }
```



Input

stdin input



Output

```
--- Final Available Inventory ---
ID: P102 -> Smartphone | Category:
Electronics | Price: $800.0 | Qty: 15
ID: P101 -> Laptop | Category: Electronics |
Price: $1200.0 | Qty: 12
```



Run



Save